

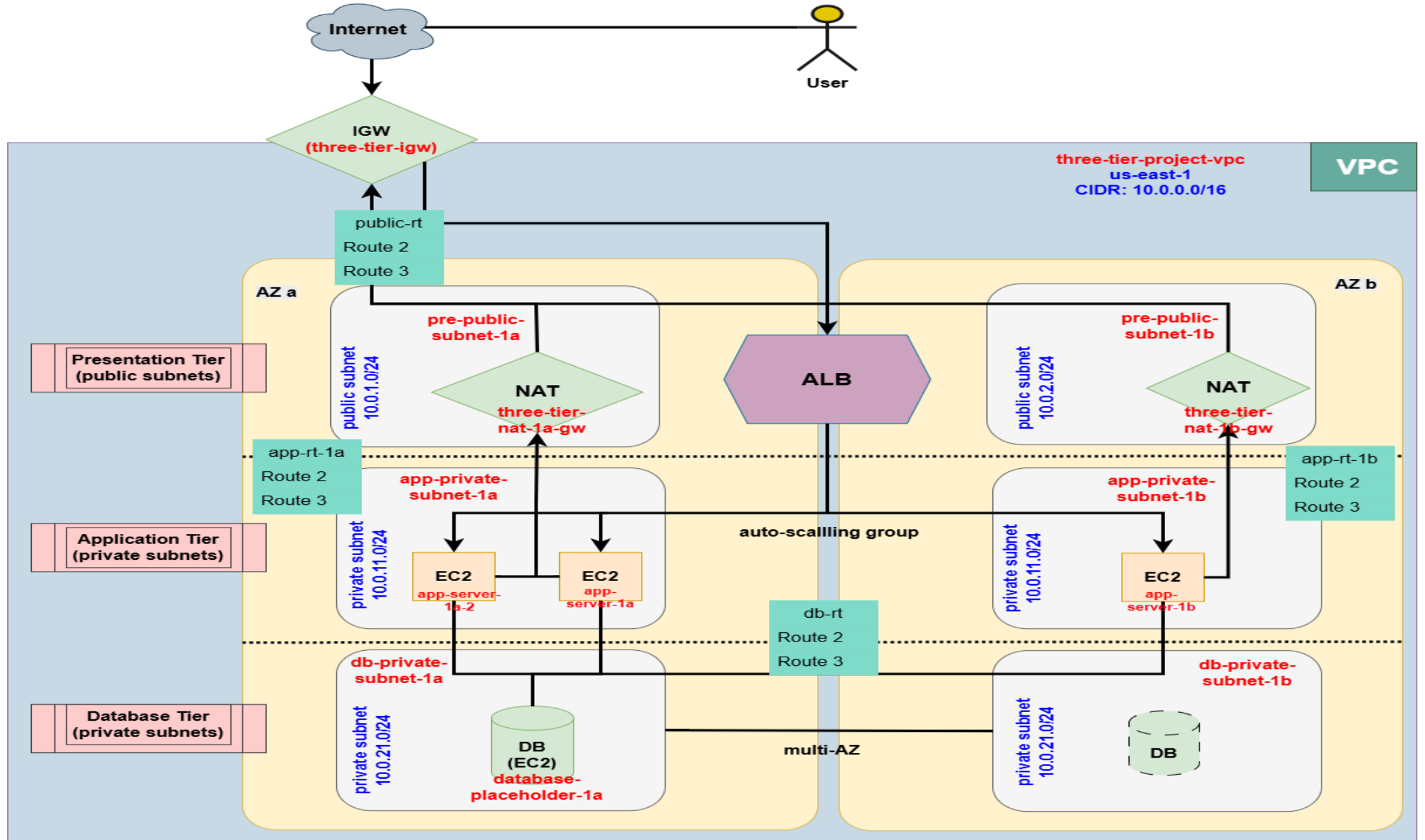
AWS 3-Tier Cloud Architecture Design & Deployment

Cloud Engineering Bootcamp



Presenter: M.Phil. Hafiz Quddus
Project -1

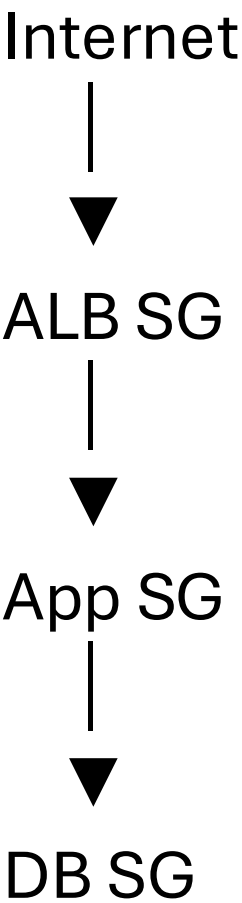
Architecture Overview



Network Security (request flow)



Security Group	Inbound Rule
ALB SG	HTTP (80)/HTTPS (443) from Internet
App Tier SG	HTTP (80) from ALB SG
Data Tier SG	Database port (3306) from App Tier SG

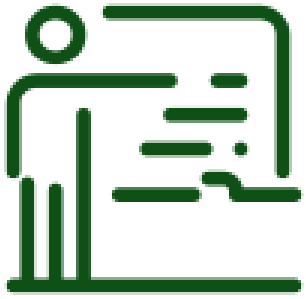


Design Decisions



Decision	Why ?
ALB	High Availability (EC2s)
Private EC2	Security
NAT Gateway	Secure outbound Internet
Session Manager	No SSH required
Health Checks	Automatic failover

Live Demo



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AWS Three-Tier Cloud Architecture

Tier 1: Presentation

Traffic received through the internet-facing Application Load Balancer.

Tier 2: Application

Instance ID: `i-076d7676f4e0a3122`
Availability Zone: `us-east-1a`

Tier 3: Data

Database status: Connected
Database endpoint: `10.0.31.199:3306`

API endpoint: </api/stats>
Health endpoint: </health>

Cost & Future Improvements (AWS Price-Calculator)

Service	Estimated Monthly cost (USD)
2 × NAT Gateways	198.90
4 × EBS (8 GB gp3) Storage	8.65
Application Load Balancer (ALB)	31.03
4 × EC2 (t3.micro)	4.16
Total	242.74

https://calculator.aws/#/estimate?utm_source=chatgpt.com&id=0436b62ddefabc1f8bb5ecaf31db50d78133a114



Service	Estimated Monthly cost (USD)
1 × NAT Gateways	99.23
4 × EBS (8 GB gp3) Storage	8.65
Application Load Balancer (ALB)	31.03
4 × EC2 (t3.micro)	4.16
Total	143.07

<https://calculator.aws/#/estimate?id=0622c06628053807d4b2ae013f85388c954cec50>

Challenges & Lesson Learned



Troubleshoots

Challenge	Solution
EC2 reboot stopped Node.js	Created a systemd service
IMDSv1 metadata issue	Switched to IMDSv2
Private EC2 access	Used Session Manager
Unhealthy targets	Fixed health checks and application startup
...	...

Summary & Q&A

Questions



Requirement

Status

VPC



Six Subnets



ALB



Private App Tier



Database Tier



Security Groups



Health Checks



High Availability



Documentation

